

Time-series lung cancer CT dataset

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Abstract—In order to better explore the evolution process of lung nodules in lung cancer patients, we collect lung CT data at multiple time points of lung cancer patients, track and mark the CT positions of the same lung nodules in lung cancer patients at different time points, and make time-series CT data sets of lung cancer patients. After that, 3D-UNet model is used to detect lung nodules on our data set. Experiment proves the effectiveness and availability of the data set, and also proved that the image data at multiple time points could improve the accuracy of the model's identification of lung nodules.

Index Terms—Time-series CT ,Medical segmentation,Lung nodule, UNet

I. INTRODUCTION

Lung cancer is one of the most common and deadly malignant cancers [1]. Although many current studies [2] have used computers to automatically segment lung nodules, many current methods focus on analyzing CT data sets of lung cancer patients at a single point in time. A large amount of time sequence information is ignored. Therefore, we continuously collect lung CT image data of lung cancer patients at multiple time points and give lung nodule markers of lung cancer patients. We make time-series CT data set of lung cancer patients, and use 3D-UNet [3] to prove that using lung cancer CT at multiple time points can improve the accuracy of lung nodule detection model.

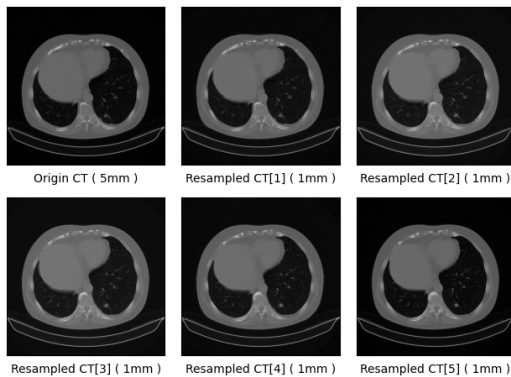


Fig. 1. Five CT images with thickness of 1mm after linear interpolation

II. DATA PROCESSING

We collect time-series CT data of 67 lung cancer patients from the hospital, and each patient has CT at 3 time points. After preprocessing the patient's original CT data, the clinician in the hospital marks the location of the lung nodules

of each lung cancer patient at each time point, and then tracks the pulmonary nodules of all lung cancer patients. The same pulmonary nodules in the CT of each patient at each time point are marked. We generate a time-series CT data set of lung cancer patients by tailoring the area around lung nodules. The following is a detailed description of the data set processing.

A. Resample

Firstly, linear interpolation algorithm is used to resample the original CT of lung cancer patients, so that the pixel size of each scanning surface and the distance between scanning surfaces of all CT of lung cancer patients is 1mm, so as to reduce errors caused by data inconsistency. Before and after the linear interpolation of the original 5mm thickness CT to 1mm thickness CT of a lung cancer patient at a certain time point are shown in Figure 1.

B. Window adjustment

In order to highlight the lesion area, make the model better learn the lesion characteristics of the lung lesion area, and exclude the interference of other organs and tissues, we adjust the window width and window level of the CT after the resampling operation .

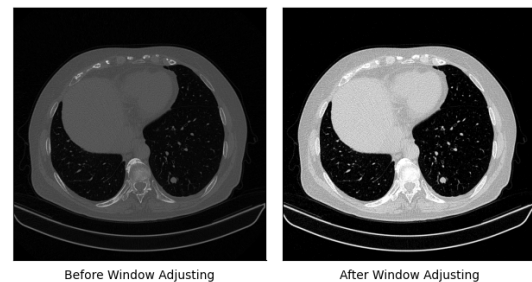


Fig. 2. CT images of lung cancer patients before and after window adjustment

C. Lung nodule markers

The doctors use the rectangle box to mark the position of lung nodules in the CT image after the above preprocessing operation. The triple imaging data of a lung cancer patient and the marked rectangle box are shown in Figure 3. Figure 3 shows continuous CT images of the same lung nodule in one patient at different time points. The duration of CT shot increases from top to bottom. The red box is the rectangular box in which the doctors identify the lung nodule.

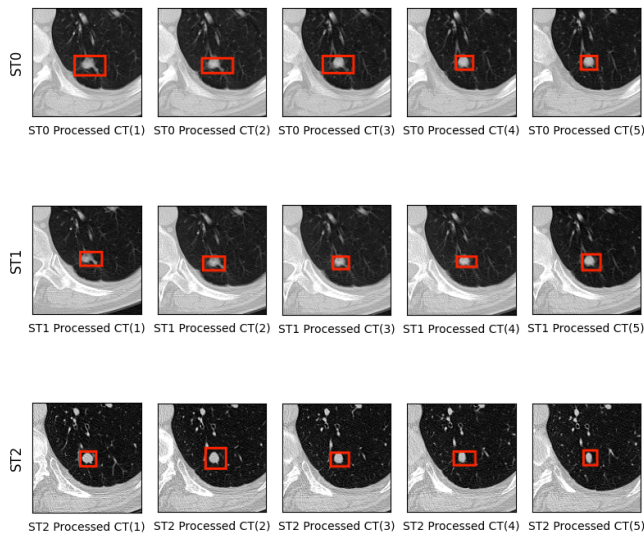


Fig. 3. Markers of lung nodules on CT

D. Fined markers

In order to mark the area of lung nodules more accurately, we make fine markers on the area of lung nodules based on the rectangular box marks given by the doctors. Figure 4 shows the original CT images of a lung cancer patient at a certain time point, the corresponding rectangular mask image of lung nodules, and the refined mask image of lung nodules. It can be seen that refined mask images can more accurately describe the position of lung nodules in CT.

III. RESULT

We randomly select 75 % of the patient data as the training set, and the remaining 25 % of the patient data as the test set. We use the third time point CT, the second and third time points CT, the first and third time points CT, and the three time

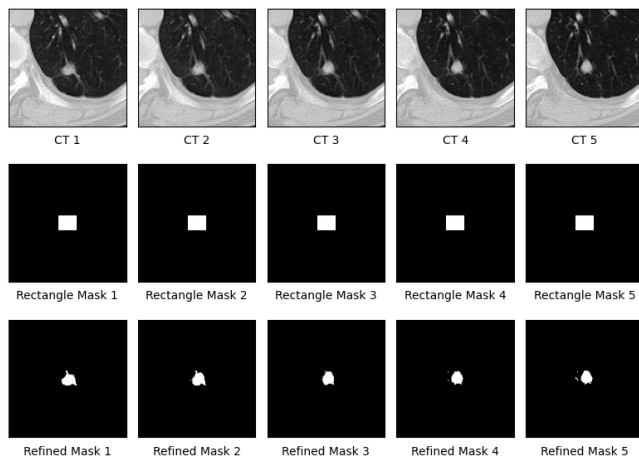


Fig. 4. Origin CT, Rectangular markers, Fined markers

TABLE I
MODEL PERFORMANCE UNDER DIFFERENT TIME POINT INPUTS

ST0	ST1	ST2	Dice	Sensitivity
		✓	0.6662	0.6424
✓		✓	0.6891	0.6952
	✓	✓	0.6626	0.6508
✓	✓	✓	0.6907	0.7210

points CT as the input of 3D-UNet. Model codes the $96 \times 128 \times 128$ CT images as a feature map of $6 \times 1024 \times 1024$ through continuous convolution operation, and samples the right half four times, gradually expands the encoded feature map to the image size of $32 \times 128 \times 128$ to detect lung nodules in the third time CT. 3D-UNet is implemented using Pytorch 1.10, and training is conducted on NVIDIA GeForce GTX 3090. Dice loss is used for training loss. The optimizer uses Adam and the learning rate is set to 0.00001. After 100 epochs of training, the model converges. The training results of the model are shown in Table 1.

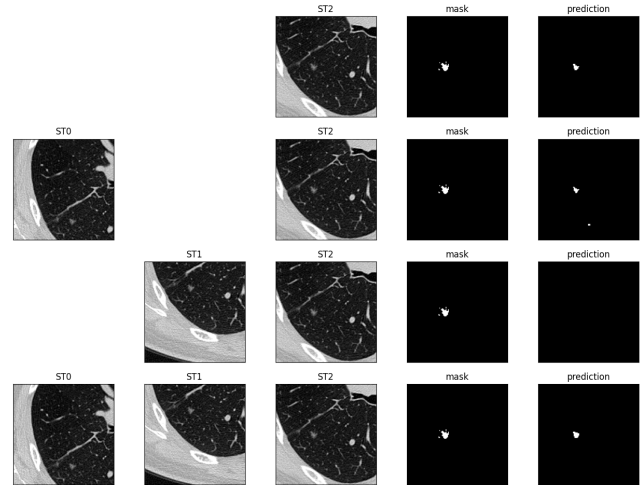


Fig. 5. Prediction at different time points

Under different time point inputs, the lung nodule results detected by the model are shown in Figure 6. It can be seen that the prediction performance of the model is the best and most accurate when multiple time points of pulmonary nodules CT are used as input, which also proves the effectiveness and availability of our data set.

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